Experiment-12

Develop a mobile application to send an email.

AIM:

To develop a mobile application that enables the user to send an email through the app. The application will provide an interface to input the recipient's email address, subject, message content, and optionally attach files. It will use Android's built-in email capabilities to send the email.

ALGORITHM:

Initialize the Android Application:

- Set up a new Android project in **Android Studio**.
- Create a **MainActivity** for the UI where the user will input email details (recipient address, subject, message, and attachment if any).

Design the User Interface (UI):

- In the **XML layout file**, create the following UI components:
  - **EditText** fields for entering:
    - Recipient's email address.
    - Subject of the email.
    - Message content.
  - **Button** to trigger the email sending process.
  - **Optional**: **Button** to select file attachments (if required).
  - **TextView** or **Toast** for displaying feedback to the user (e.g., success or failure).

Declare Variables:

- Declare variables for **EditText** fields (for email address, subject, and message).
- Declare a **Button** for sending the email and another **Button** for attachment (optional).
- Declare a **String** variable to hold the recipient email, subject, and message content.

Set OnClickListener for Send Button:

- Set an **OnClickListener** for the **Send Email Button**.

- Inside the onClick() method, retrieve the values entered by the user:
    - Get the email recipient, subject, and message using getText().toString() for the **EditText** fields.

Validate User Input:

- Ensure that the **email address** is in a valid format (e.g., name@domain.com).
- Check if all necessary fields (email, subject, message) are not empty.
- If any validation fails, show an error message (Toast) and do not proceed.

Create an Intent to Send the Email:

- Use an **Intent** to trigger the **email client** (e.g., Gmail, Outlook, etc.).
    - Set the action to Intent.ACTION_SENDTO.
    - Use mailto: URI scheme to trigger the email client and set the **recipient, subject, and message** as parameters.
    - Example: mailto:recipient@example.com?subject=Subject&body=Message
- Set the **email fields** (recipient, subject, message) in the intent.

Optional: Attach Files (if applicable):

- If attachments are required, use Intent.EXTRA_STREAM to attach a file (e.g., an image or document).
- Use a file picker to allow the user to select the attachment, then add the file URI to the intent.

Send the Email:

- Call startActivity(intent) to invoke the email client and send the email.
- If the email client is installed, it will open with the email details prefilled, and the user can press the send button.
- If the email client is not installed or an error occurs, display an appropriate **Toast message** informing the user.

Display Feedback:

- If the email was successfully sent, show a **Toast** or update the UI with a success message.
- If there was an error (e.g., invalid email, missing email client), display an error **Toast** or show an error message in the **TextView**.

End the Operation:

- After the email has been sent or if there was an error, the operation is complete. The user can send another email or exit the app.

CODE:

activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:autofillHints=""
        android:hint="@string/recipient_email"
        android:inputType="textEmailAddress"
        android:minHeight="48dp"
        android:textColorHint="#546E7A"
        tools:ignore="VisualLintTextFieldSize" />

    <EditText
        android:id="@+id/subject"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:autofillHints=""
        android:hint="@string/subject"
        android:inputType="text"
        android:minHeight="48dp"
        android:textColorHint="#546E7A"
        tools:ignore="VisualLintTextFieldSize" />

    <EditText
        android:id="@+id/message"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:autofillHints=""
        android:hint="@string/message"
        android:inputType="text"
        android:lines="5"
        android:minHeight="48dp"
        android:textColorHint="#546E7A"
        tools:ignore="VisualLintTextFieldSize" />

    <Button
        android:id="@+id/sendBtn"
        android:text="@string/send_email"
```

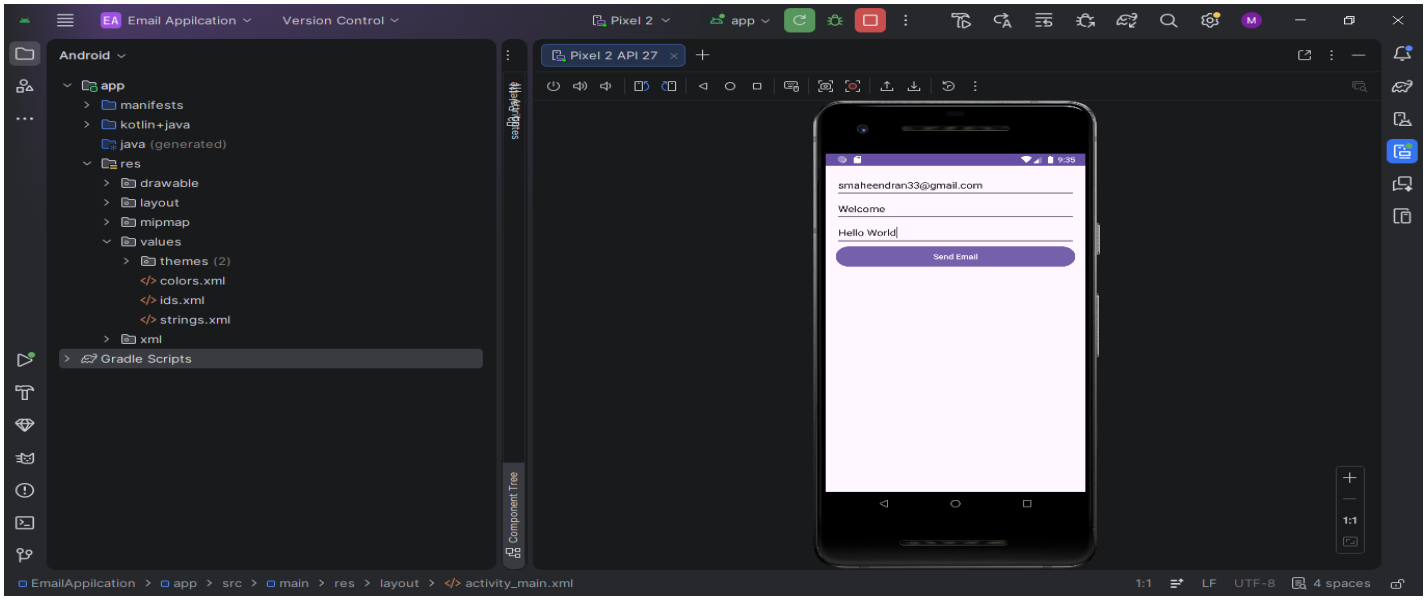
```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        tools:ignore="VisualLintButtonSize" />
</LinearLayout>
```

MainActivity.java:

```
package com.example.emailappilcation;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText email, subject, message;
    Button sendBtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        email = findViewById(R.id.email);
        subject = findViewById(R.id.subject);
        message = findViewById(R.id.message);
        sendBtn = findViewById(R.id.sendBtn);

        sendBtn.setOnClickListener(v -> {

            String to = email.getText().toString().trim();
            String sub = subject.getText().toString().trim();
            String msg = message.getText().toString().trim();

            if (to.isEmpty() || sub.isEmpty() || msg.isEmpty()) {
                Toast.makeText(MainActivity.this,
                        "Please fill all fields",
                        Toast.LENGTH_SHORT).show();
                return;
            }

            Intent intent = new Intent(Intent.ACTION_SEND);
            intent.setType("message/rfc822");
            intent.putExtra(Intent.EXTRA_EMAIL, new String[]{to});
            intent.putExtra(Intent.EXTRA_SUBJECT, sub);
            intent.putExtra(Intent.EXTRA_TEXT, msg);

            try {
                startActivity(Intent.createChooser(intent, "Send Email"));

                // □ Success message (after opening mail app)
```

```
                Toast.makeText(MainActivity.this,
                        "Email app opened successfully",
                        Toast.LENGTH_SHORT).show();

            } catch (Exception e) {
                Toast.makeText(MainActivity.this,
                        "No email app found",
                        Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

OUTPUT:



RESULT:

A mobile application is developed to send an email.